L0.10 — Convolution Theorem Deep Dive

1. Start with what we already know

From convolution:

$$y(t) = x(t) * h(t)$$

which means:

$$\boxed{y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau}$$

From Fourier:

$$X(\omega) = \int x(t)e^{-j\omega t}dt$$

$$H(\omega) = \int h(t)e^{-j\omega t}dt$$

and:

$$Y(\omega) = \int y(t)e^{-j\omega t}dt.$$

We want to discover the relationship between:

$$Y(\omega)$$

and:

$$X(\omega), H(\omega).$$

---

2. What would we hope happens?

Spatial/time domain convolution is expensive conceptually:

```
flip
shift
multiply
integrate
repeat
```

If somehow Fourier could transform this into:

$$X(\omega)H(\omega),$$

then filtering would become:

```
transform input
×
transform filter
→
inverse transform
```


That would be enormously simpler mathematically.

And that's exactly what happens.

---

3. Let's derive it carefully

Start with:

$$y(t) = x * h.$$

Therefore:

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau.$$

Take the Fourier transform:

$$Y(\omega) = \int_{-\infty}^{\infty} y(t)e^{-j\omega t}dt.$$

Substitute $y(t)$:

$$Y(\omega) = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau\right] e^{-j\omega t}dt.$$

That looks horrible.

But we'll simplify it.

---

4. Rearrange the integrals

Under suitable conditions, interchange integration:

$$Y(\omega) = \int_{-\infty}^{\infty} x(\tau)\left[\int_{-\infty}^{\infty} h(t-\tau)e^{-j\omega t}dt\right]d\tau.$$

Focus only on the inner integral:

$$\int h(t-\tau)e^{-j\omega t}dt.$$

Now make a substitution.

Let:

$$u = t - \tau.$$

Then:

$$t = u + \tau.$$

Also:

$$dt = du.$$

So:

$$\int h(u)e^{-j\omega(u+\tau)}du.$$

Split the exponential:

$$e^{-j\omega(u+\tau)} = e^{-j\omega u}e^{-j\omega\tau}.$$

Therefore:

$$= e^{-j\omega\tau}\int h(u)e^{-j\omega u}du.$$

But:

$$\int h(u)e^{-j\omega u}du = H(\omega).$$

So inner integral becomes:

$$\boxed{e^{-j\omega\tau}H(\omega)}.$$

---

5. Put it back into the outer integral

Now:

$$Y(\omega) = \int x(\tau)e^{-j\omega\tau}H(\omega)d\tau.$$

Since $H(\omega)$ does not depend on τ:

$$Y(\omega) = H(\omega)\int x(\tau)e^{-j\omega\tau}d\tau.$$

But:

$$\int x(\tau)e^{-j\omega\tau}d\tau = X(\omega).$$

Therefore:

$$\boxed{Y(\omega) = X(\omega)H(\omega)}$$

and since:

$$y = x * h,$$

we get:

$$\boxed{x * h \quad \longleftrightarrow \quad XH}$$

Boom.

That is the convolution theorem.

---

6. What just happened conceptually?

In the time/spatial domain:

$$x$$

interacts with shifted copies of:

$$h.$$

In the Fourier domain, those complicated interactions become independent multiplications at each frequency.

So:

$$\boxed{\text{convolution couples neighboring positions}}$$

but:

$$\boxed{\text{Fourier diagonalizes the filtering operation}}$$

That last sentence is deeper linear algebra.

You don't need to memorize the word “diagonalizes” yet, but here's the intuition:

> Fourier changes the representation so that the filter no longer mixes neighboring samples; it simply scales each frequency independently.

That is huge.

---

7. Think of the filter as knobs

Suppose the signal has frequency components:

$$X(\omega_1), X(\omega_2), X(\omega_3).$$

The filter has:

$$H(\omega_1), H(\omega_2), H(\omega_3).$$

Then:

$$Y(\omega_1) = X(\omega_1)H(\omega_1)$$

$$Y(\omega_2) = X(\omega_2)H(\omega_2)$$

$$Y(\omega_3) = X(\omega_3)H(\omega_3).$$

Each frequency gets its own multiplier.

So $H(\omega)$ is like:

```
frequency 1 → gain
frequency 2 → gain
frequency 3 → gain
...
```

This is exactly what a filter does in frequency language.

---

8. Example

Suppose signal spectrum:

$$X = [10, 8, 4, 2]$$

for four conceptual frequency components.

Suppose filter:

$$H = [1, 1, 0.5, 0.1].$$

Then:

$$Y = XH$$

elementwise:

$$Y = [10(1), 8(1), 4(0.5), 2(0.1)].$$

Therefore:

$$\boxed{Y = [10, 8, 2, 0.2].}$$

Interpretation:

- low frequencies mostly preserved;
- higher ones attenuated.

That is a low-pass behavior.

---

9. Very important: this is element-wise multiplication

Not matrix multiplication.

At each frequency:

$$\boxed{Y(\omega) = X(\omega) \times H(\omega).}$$

Later for DFT:

$$Y[u] = X[u]H[u].$$

And for 2-D images:

$$\boxed{G[u, v] = F[u, v]H[u, v].}$$

Every frequency bin gets multiplied by the corresponding filter value.

---

10. Now connect to blur

Remember Gaussian-like smoothing:

$$h = \frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}.$$

Spatial interpretation:

> neighboring pixels are averaged with larger weights near the centre.

Frequency interpretation:

> lower spatial frequencies pass strongly; higher spatial frequencies are attenuated.

So:

$$\boxed{\text{Gaussian blur kernel } h}$$

and:

$$\boxed{\text{Gaussian-like low-pass frequency response } H}$$

are two descriptions of the same filter.

---

11. One filter, two domains

You should now think:

$$\boxed{h \leftrightarrow H}$$

where:

h

tells:

> What happens to an impulse?

H

tells:

> What happens to each frequency?

This is an extremely important pair.

---

12. Example: impulse-response viewpoint

Suppose:

$$h[n] = \frac{1}{3}[1, 1, 1].$$

Impulse response says:

> One impulse spreads equally over three adjacent positions.

That's smoothing.

Frequency response says:

> Some frequencies pass strongly, rapidly alternating frequencies are attenuated.

Same system.

---

13. Let's prove that intuition for one frequency

Take input:

$$x[n] = e^{j\omega n}.$$

Convolve:

$$y[n] = \sum_k h[k]x[n-k].$$

Substitute:

$$x[n-k] = e^{j\omega(n-k)}.$$

Then:

$$y[n] = \sum_k h[k]e^{j\omega(n-k)}.$$

Factor:

$$e^{j\omega n}$$

out:

$$y[n] = e^{j\omega n}\sum_k h[k]e^{-j\omega k}.$$

The sum:

$$\sum_k h[k]e^{-j\omega k}$$

is the frequency response:

$$H(\omega).$$

Therefore:

$$\boxed{y[n] = H(\omega)e^{j\omega n}}$$

Again:

$$\boxed{\text{same frequency in} \rightarrow \text{same frequency out}}$$

only scaled/phase-shifted by:

$$H(\omega).$$

This is the discrete counterpart of what we saw in L0.8.

---

14. Example with averaging filter

Take:

$$h[n] = \frac{1}{2}[1, 1].$$

Its frequency response:

$$H(\omega) = \frac{1}{2}\left(1 + e^{-j\omega}\right).$$

Let's examine particular frequencies.

---

15. Frequency $\omega = 0$

Then:

$$e^{-j0} = 1.$$

So:

$$H(0) = \frac{1}{2}(1+1) = \boxed{1}.$$

DC passes unchanged.

That's what we expect from a normalized average filter.

---

16. Frequency $\omega = \pi$

Now:

$$e^{-j\pi} = -1.$$

Thus:

$$H(\pi) = \frac{1}{2}(1-1) = \boxed{0}.$$

What does $\omega = \pi$ correspond to in discrete signals?

The maximally alternating pattern:

$$[1, -1, 1, -1, \ldots].$$

Let's apply the average directly.

For pairs:

$$\frac{1}{2}(1 + (-1)) = 0.$$

Exactly.

So averaging completely removes this very high-frequency alternating pattern.

Beautiful.

---

17. This is the real meaning of low-pass filtering

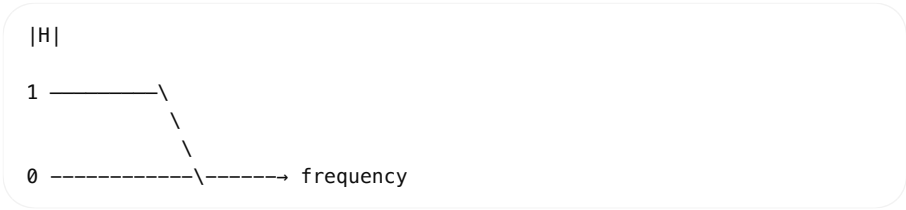
A low-pass filter has:

$$|H(\omega)|$$

large for low ω,

and small for high ω.

Conceptually:



It “passes” low frequencies.

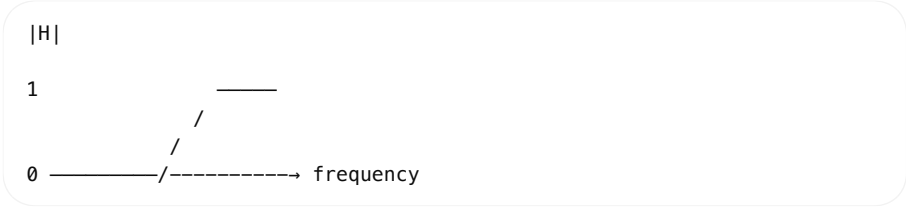
Hence:

$$\boxed{\text{low-pass filter}}.$$

---

18. High-pass filter

A high-pass response looks conceptually:



Low frequencies suppressed.

High frequencies preserved/emphasized.

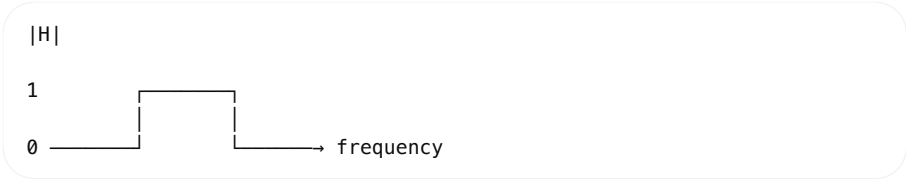
Useful for:

- edges;
- detail;
- sharpening.

---

19. Band-pass filter

Pass only intermediate frequencies:

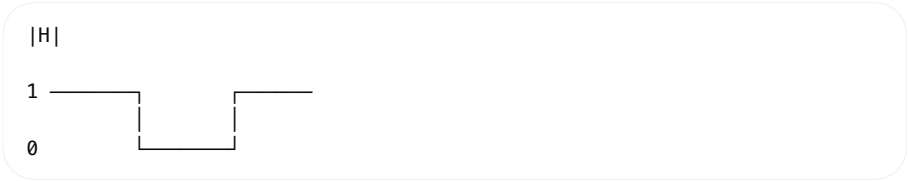


Can isolate specific texture/repetition scales.

---

20. Band-stop / notch filter

Suppress only a frequency band:



Useful for periodic interference.

Later in images, **notch filtering** is often used to remove periodic noise.

---

21. Why periodic noise is easy to attack in Fourier domain

Suppose image contains unwanted repeated stripes.

Spatially:

```
~ ~ ~ ~ ~ ~ ~ ~
```

That pattern is spread across the whole image.

Hard to remove locally.

But because it is periodic, its energy may concentrate at specific frequency locations.

So in Fourier domain:

```
mostly normal spectrum
+
a few bright unwanted peaks
```

We can suppress those peaks.

Then inverse Fourier transform.

That's often much more elegant than designing a local spatial filter.

---

22. Now connect sharpening

Suppose original image spectrum:

$$F(u,v).$$

A sharpening filter might make:

$$|H(u,v)|$$

larger at higher frequencies.

Then:

$$G = FH$$

amplifies fine changes.

So:

$$\boxed{\text{sharpening} \approx \text{high-frequency emphasis}}$$

conceptually.

---

23. Derivative filters and frequency

Remember discrete derivative:

$$[-1, 1].$$

It kills constants:

$$\sum h = 0.$$

So:

$$H(0) = 0.$$

That means:

$$\boxed{\text{DC is eliminated.}}$$

And it responds more strongly as frequency increases.

This is why derivatives behave like high-pass operations.

Again:

$$\boxed{\text{spatial derivative} \leftrightarrow \text{high-frequency emphasis.}}$$

---

24. Important relationship with kernel sum

For a discrete filter:

$$H(0) = \sum_k h[k].$$

Why?

Frequency response:

$$H(\omega) = \sum_k h[k] e^{-j\omega k}.$$

At:

$$\omega = 0,$$

all exponentials become 1:

$$H(0) = \sum_k h[k].$$

Therefore:

If

$$\sum h = 1,$$

then:

$$H(0) = 1.$$

DC preserved.

If

$$\sum h = 0,$$

then:

$$H(0) = 0.$$

DC removed.

This mathematically confirms the kernel-sum intuition we learned in L0.4/L0.6.

That is a fantastic connection.

---

25. Mean filter example

Mean kernel:

$$h = \frac{1}{3}[1, 1, 1].$$

Sum:

$$1.$$

Therefore:

$$H(0) = 1.$$

Constant signal survives.

Derivative:

$$h = [-1, 1].$$

Sum:

$$0.$$

Therefore:

$$H(0) = 0.$$

Constant signal disappears.

Now these facts aren't merely heuristics anymore.

They're frequency-response statements.

---

26. Why convolution theorem is more than speed

There are two benefits.

Benefit 1 — understanding

Spatial domain may tell you:

> This kernel averages neighbors.

Frequency domain tells you:

> This kernel suppresses high spatial frequencies.

Sometimes one description is much more intuitive.

Benefit 2 — computation

For long signals or large kernels:

$$x * h$$

can be expensive.

Instead:

$$x \to X$$

$$h \to H$$

then:

$$Y = XH$$

then inverse transform:

$$y = \mathrm{F}^{-1}\{Y\}.$$

Your lecture uses exactly this route to motivate FFT convolution.

---

27. Spatial-domain filtering pipeline

Direct approach:

```
input x
  ↓
convolve with h
  ↓
output y
```

Mathematically:

$$\boxed{y = x * h.}$$

---

28. Frequency-domain filtering pipeline

Alternative:

```
x ──FT──> X
          │
          × H
          │
          ▼
          Y
          │
       inverse FT
          ↓
          y
```

Mathematically:

$$\boxed{y = \mathrm{F}^{-1}\{XH\}.}$$

This equation should start feeling very natural now.

---

29. Tiny conceptual example

Suppose signal consists of:

$$x(t) = \cos(t) + \cos(10t).$$

Two frequencies:

$$1$$

and:

$$10.$$

Suppose filter:

$$H(1) = 1$$

$$H(10) = 0.$$

Then:

$$Y(1) = X(1)$$

and:

$$Y(10) = 0.$$

Inverse transform yields:

$$\boxed{y(t) = \cos(t).}$$

We removed the high-frequency component.

Spatial/time domain convolution did the same thing, but frequency interpretation makes the operation transparent.

---

30. Another example — remove DC

Suppose:

$$x(t) = 5 + \cos(4t).$$

So:

- DC = 5;
- oscillatory component = frequency 4.

Suppose:

$$H(0) = 0$$

and:

$$H(4) = 1.$$

Then DC disappears.

Output:

$$\boxed{y(t) = \cos(4t).}$$

A filter with zero DC response removes constant offset/background level.

This is related to high-pass/derivative behavior.

31. 2-D convolution theorem

For images:

$$g[m,n] = f[m,n] * h[m,n].$$

In the 2-D frequency domain:

$$\boxed{G(u,v) = F(u,v)H(u,v).}$$

Same theorem.

Now u, v correspond to horizontal and vertical spatial frequencies.

This will be central later.

---

32. Example: Gaussian blur in 2-D

Spatial kernel:

$$h[m,n]$$

is Gaussian-like.

Its frequency response:

$$H(u,v)$$

is also smooth and low-pass.

Low frequencies near DC:

$$H \approx 1.$$

Farther high frequencies:

$$H \to 0.$$

Thus:

$$G = FH$$

keeps central low-frequency content and suppresses outer high-frequency content.

Result:

$$\boxed{\text{blurred image}}.$$

---

33. Why edge information appears farther from DC

In a centered 2-D Fourier visualization:

```
center
↓
low frequencies

farther from center
↓
higher frequencies
```

So smooth structure concentrates toward center.

Fine detail spreads farther outward.

This becomes visually obvious once we learn `fftshift`.

Not yet — that's DFT territory.

---

34. Important distinction: Fourier transform of kernel vs kernel itself

Don't confuse:

$$h$$

with:

$$H.$$

h:

```
spatial coefficients
```

for example:

$$\frac{1}{3}[1,1,1].$$

H:

```
frequency response of those coefficients
```

describes their effect across frequencies.

The Fourier transform converts between them:

$$\boxed{h \leftrightarrow H.}$$

---

35. What if $H(\omega) = 1$ for all frequencies?

Then:

$$Y = X.$$

Therefore:

$$y = x.$$

What kernel produces this?

The impulse:

$$h(t) = \delta(t).$$

Because:

$$\delta(t) \leftrightarrow 1.$$

So:

$$\boxed{\text{identity in space} \leftrightarrow \text{unity gain at every frequency.}}$$

Beautiful.

---

36. What if $H(\omega) = 0$ everywhere?

Then:

$$Y = 0.$$

Output:

$$y = 0.$$

The system suppresses everything.

---

37. What if $H(0) = 1$ but $H(\omega) = 0$ elsewhere?

Only DC survives.

Output becomes constant equal to some average component.

That's an extreme low-pass filter.

An image would become essentially a uniform gray image corresponding to average brightness.

---

38. What if $H(0) = 0$ but high frequencies pass?

Constant areas disappear.

Edges/details remain.

That's a high-pass-like output.

Again:

$$\boxed{\text{frequency response tells us filter behavior instantly.}}$$

---

39. Convolution theorem has a dual partner

There is another property:

$$\boxed{x(t)h(t) \leftrightarrow \frac{1}{2\pi} X * H}$$

depending on transform convention.

Meaning:

- convolution in one domain;
- multiplication in the other.

This is called **duality between multiplication and convolution**.

Do not memorize the constants yet.

Your quiz priority right now is:

$$\boxed{x * h \leftrightarrow XH.}$$

---

40. Why this matters for windowing

Later when you multiply a signal by a finite window:

$$x(t)w(t),$$

in frequency domain you convolve:

$$X * W.$$

That can cause spectral spreading/leakage.

This will matter later with DFT/windowing.

Not for today — just notice how deep the theorem reaches.

---

41. Important subtlety: continuous theorem vs DFT theorem

For ordinary continuous Fourier transform:

$$\boxed{x * h \leftrightarrow XH.}$$

For finite DFT:

$$\boxed{\text{DFT multiplication naturally corresponds to circular convolution.}}$$

Your lecture explicitly warns about this.

This difference is the reason we'll need:

$$\boxed{\text{zero padding}}$$

later.

But we're not solving circular convolution yet.

That's L0.12.

---

42. Important exam/quiz distinction

If asked:

> What does multiplication of DFTs correspond to?

Do **not** automatically say ordinary linear convolution.

Strictly:

$$\boxed{\text{circular convolution}}$$

unless signals have been appropriately zero-padded.

This is an important trap.

We'll master it later.

---

43. Worked example — infer filter type from H

Suppose:

$$|H(\omega)| = \begin{cases} 1, & |\omega| < 5 \\ 0, & |\omega| > 5. \end{cases}$$

What kind of filter?

Low frequencies:

$$\text{pass.}$$

High frequencies:

$$\text{stop.}$$

Therefore:

$$\boxed{\text{low-pass filter}}.$$

---

44. Worked example — high-pass

$$|H(\omega)| = \begin{cases} 0, & |\omega| < 5 \\ 1, & |\omega| > 5. \end{cases}$$

Then:

$$\boxed{\text{high-pass filter}}.$$

For an image, likely emphasizes rapid intensity changes.

---

45. Worked example — notch

Suppose:

$$H(\omega) = 1$$

everywhere except:

$$\omega = \pm 20,$$

where:

$$H = 0.$$

This suppresses only the frequency pair at ± 20.

This is a:

$$\boxed{\text{notch filter}}.$$

Potential use:

$$\boxed{\text{remove periodic interference at that frequency}}.$$

---

46. Worked example using frequency components

Signal:

$$x(t) = 3 + 4\cos(2t) + 2\cos(10t).$$

Filter:

$$H(0) = 1,$$

$$H(2) = 0.5,$$

$$H(10) = 0.$$

Ignoring phase modifications for simplicity:

DC becomes:

$$3(1) = 3.$$

Frequency 2:

$$4(0.5) = 2.$$

Frequency 10:

$$2(0) = 0.$$

So:

$$\boxed{y(t) = 3 + 2\cos(2t).}$$

Interpretation:

high-frequency variation removed.

---

47. Worked example from a kernel

Take:

$$h[n] = \frac{1}{2}[1, 1].$$

Input:

$$x[n] = (-1)^n.$$

That sequence is:

$$[1, -1, 1, -1, \ldots].$$

Direct filtering:

$$y[n] = \frac{1}{2}\left[x[n] + x[n-1]\right].$$

Since neighbors are opposites:

$$1 + (-1) = 0.$$

Thus:

$$\boxed{y[n] = 0}.$$

Frequency interpretation:

this input is the highest discrete-time oscillation around $\omega = \pi$.

And earlier:

$$H(\pi) = 0.$$

Same result.

This is exactly the power of having two representations.

---

48. Challenge — identify filter from kernel sum

Kernel A

$$h_A = \frac{1}{4}[1, 2, 1].$$

Sum:

$$1.$$

Likely preserves:

$$\boxed{\text{DC}}.$$

Likely behavior:

$$\boxed{\text{smoothing / low-pass}}.$$

Kernel B

$$h_B = [-1, 0, 1].$$

Sum:

$$0.$$

Therefore:

$$H_B(0) = 0.$$

Likely behavior:

$$\boxed{\text{derivative/high-pass-like}}.$$

Great shortcut.

---

49. Challenge — physical camera interpretation

A camera PSF is very broad.

What does that suggest about its frequency response?

A broad PSF means strong spatial spreading/blurring.

That usually corresponds to a **narrower low-pass frequency response**:

$$\boxed{\text{high frequencies strongly attenuated.}}$$

Therefore fine detail is lost.

---

50. Opposite camera interpretation

An ideal PSF is approximately:

$$\delta(x, y).$$

Its transform is:

$$H(u, v) = 1.$$

So all frequencies pass.

Therefore:

$$\boxed{\text{perfect spatial impulse response} \leftrightarrow \text{perfect frequency transmission.}}$$

Real optical systems cannot maintain this perfectly.

This connects frequency response to optical resolution.

---

51. One important imaging concept: MTF preview

Later, imaging systems often describe resolution using something called the:

$$\boxed{\text{Modulation Transfer Function (MTF)}}.$$

Conceptually, MTF tells:

> How much contrast does the imaging system preserve at different spatial frequencies?

This is related to the magnitude of an optical transfer function, which itself is connected to the Fourier transform of the PSF.

We do not need MTF mathematically yet.

But now you already have the foundation to understand it later:

$$\boxed{\text{PSF} \overset{\text{F}}{\longleftrightarrow} \text{frequency response}}$$

and frequency-response magnitude tells how detail of different spatial scales survives.

---

52. Why a blurry imaging system loses fine details

Fine detail:

$$\rightarrow \text{higher spatial frequencies.}$$

If:

$$|H(u, v)|$$

becomes very small there:

$$F(u, v)H(u, v) \approx 0.$$

Those frequency components vanish from the measured image.

This isn't merely “pixels got averaged.”

It means actual frequency information was suppressed.

That perspective becomes essential for restoration.

---

53. Why restoration can be impossible

Suppose:

$$H(\omega) = 0$$

at some frequency.

Then:

$$Y(\omega) = X(\omega)H(\omega) = 0.$$

You observe zero.

Can you uniquely determine what:

$$X(\omega)$$

was?

No.

Because:

$$5(0) = 0$$

$$100(0) = 0$$

$$-20(0) = 0.$$

Information has been destroyed at that frequency.

This is one reason deblurring cannot always perfectly recover an image.

Great connection to Week 9 restoration.

---

54. Add noise

Real image degradation:

$$g = f * h + \eta.$$

Fourier:

$$\boxed{G = FH + N.}$$

Naive inversion:

$$F = \frac{G}{H}.$$

But:

$$G = FH + N.$$

Therefore:

$$\frac{G}{H} = F + \frac{N}{H}.$$

If:

$$H$$

is tiny:

$$\frac{N}{H}$$

can become enormous.

Noise gets amplified.

That's the basic reason simple inverse filtering is unstable.

Wiener filtering later tries to handle this intelligently.

You already have the conceptual seed.

---

55. The whole course is starting to connect

```
Impulse
   ↓
Impulse response h
   ↓
Convolution
   ↓
Image blur / filtering
   ↓
Fourier transform
   ↓
Frequency response H
   ↓
Convolution theorem
   ↓
G = FH
   ↓
Filtering / restoration / FFT
```

This is why Lecture 0 is genuinely foundational.